Supplementary Information

Physics-Aware Lightweight Segmentation for Quantitative Holographic Cell Analysis with Optical Mass Preservation

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S-I. LoRA Rank Sweep for All Architectures

Table S-1 lists the segmentation and integrated-phase results of the full objective for every architecture and LoRA rank (r = 2, 4, 8, 16, and 32), and Fig. S-1 shows every training seed. The main text uses EdgeSAM (r=32) as the primary configuration and MobileSAM (r=8) as the sensitivity comparison (shaded rows); this choice is descriptive and not the result of a selection rule. The dependence on rank was not monotonic for any architecture. Training was unstable in some seeds (Fig. S-2): EdgeSAM lost the spherocyte class in seed 44 at r = 2, 8, and 16, MobileSAM (r=32) collapsed in seed 43 in the second epoch and lost two classes, and MobileNet-UNet lost at least one class in every seed at every rank. Latency and memory were measured for the rank-8 models only (Table 4 of the main text).

TABLE S-1 LoRA rank sweep, full objective. Mean ± SD over seeds 42, 43, and 44; shaded: configurations used in the main text.

Architecture	LoRA rank	Trainable params	Mean Dice	AJI	Boundary F1	Global IPE (%)	Seeds with a lost class (mask values)
EdgeSAM	2	0.30 M (5.2%)	0.625 ± 0.108	0.249 ± 0.017	0.598 ± 0.020	4.45 ± 2.71	1/3 (seed 44: 2)
	4	0.38 M (6.5%)	0.703 ± 0.009	0.252 ± 0.021	0.615 ± 0.012	3.48 ± 0.91	0/3
	8	0.53 M (8.9%)	0.647 ± 0.114	0.249 ± 0.043	0.608 ± 0.009	3.81 ± 1.82	1/3 (seed 44: 2)
	16	0.85 M (13.4%)	0.665 ± 0.114	0.277 ± 0.022	0.623 ± 0.024	4.66 ± 3.53	1/3 (seed 44: 2)
	32	1.47 M (21.2%)	0.738 ± 0.006	0.335 ± 0.020	0.642 ± 0.005	3.02 ± 0.84	0/3
MobileSAM	2	0.63 M (9.4%)	0.808 ± 0.003	0.264 ± 0.013	0.801 ± 0.006	1.30 ± 0.09	0/3
	4	0.64 M (9.6%)	0.811 ± 0.004	0.268 ± 0.037	0.803 ± 0.002	1.34 ± 0.07	0/3
	8	0.67 M (10.0%)	0.810 ± 0.006	0.262 ± 0.017	0.802 ± 0.011	1.32 ± 0.09	0/3
	16	0.73 M (10.8%)	0.813 ± 0.002	0.279 ± 0.054	0.801 ± 0.013	1.29 ± 0.18	0/3
	32	0.85 M (12.3%)	0.687 ± 0.234	0.283 ± 0.020	0.794 ± 0.023	1.76 ± 0.53	1/3 (seed 43: 2, 4)
MobileNet-UNet	2	5.47 M (71.4%)	0.410 ± 0.004	0.534 ± 0.029	0.816 ± 0.006	2.61 ± 0.23	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	4	5.51 M (71.6%)	0.406 ± 0.002	0.560 ± 0.035	0.805 ± 0.006	2.50 ± 0.19	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	8	5.58 M (71.8%)	0.412 ± 0.004	0.551 ± 0.053	0.814 ± 0.003	2.60 ± 0.07	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	16	5.73 M (72.4%)	0.414 ± 0.002	0.505 ± 0.012	0.805 ± 0.009	3.09 ± 0.65	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	32	6.04 M (73.4%)	0.476 ± 0.108	0.541 ± 0.049	0.824 ± 0.014	5.83 ± 5.75	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2)

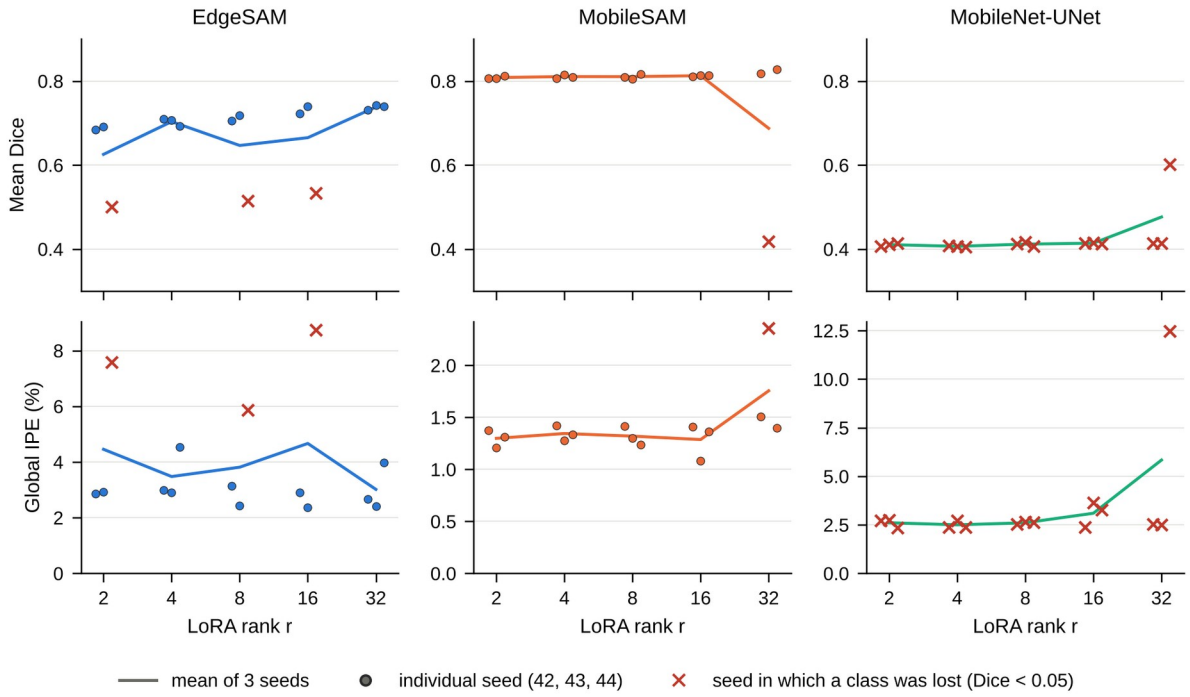


Fig. S-1 Mean Dice (top) and global IPE (bottom) against LoRA rank, full objective. Dots: individual seeds; crosses: seeds in which a class was lost (Dice < 0.05); line: mean of three seeds.

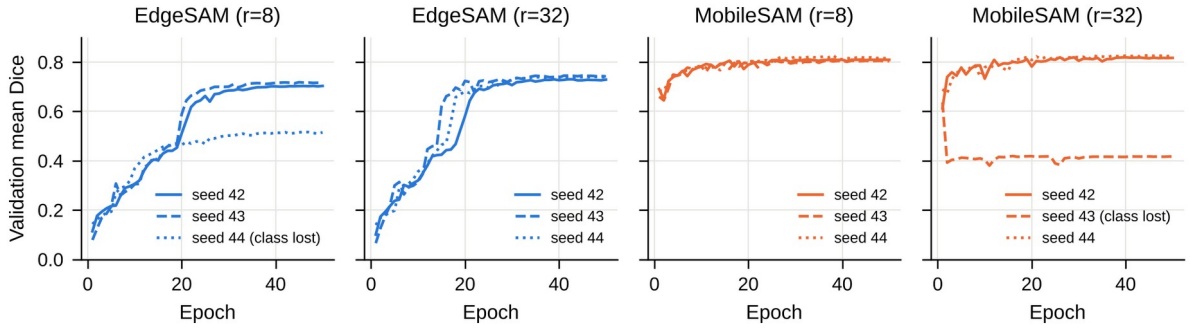


Fig. S-2 Validation mean Dice during training of EdgeSAM and MobileSAM at r=8 and r=32, one line per seed; seeds that ended with a lost class are marked in the legend.

S-II. Loss Ablation at All LoRA Ranks

Table S-2 gives the component-wise ablation (D: Dice only; P: Dice with PMC; B: Dice with PMC and BGA; F: full objective with IPP) for every architecture and rank. Over the 15 architecture–rank configurations, the mean global IPE of the full objective was lower than that of Dice alone in 15 and lower than that of Dice with PMC and BGA in 13; the seed-paired 95% interval excluded zero in 3 and 1 configurations, respectively. Over the 45 seed-paired comparisons, the full objective had the lower IPE in 42 against Dice alone and in 36 against Dice with PMC and BGA. A class was lost in 6, 3, and 13 of the 15 Dice-only runs of EdgeSAM, MobileSAM, and MobileNet-UNet, and in 3, 1, and 15 of the 15 full-objective runs. For EdgeSAM, the AJI of the full objective was higher than that of Dice with PMC and BGA in 0 of 15 seed-paired comparisons, and its Dice was higher than with Dice alone in 9 of 15.

TABLE S-2 Loss ablation at all LoRA ranks (means over seeds 42, 43, and 44).

Architecture	LoRA rank	Mean Dice (D / P / B / F)	Global IPE, % (D / P / B / F)	Seeds with a lost class (D / P / B / F)
EdgeSAM	2	0.576 / 0.710 / 0.717 / 0.625	14.42 / 5.26 / 4.88 / 4.45	0 / 0 / 0 / 1
	4	0.610 / 0.727 / 0.723 / 0.703	10.96 / 4.68 / 4.98 / 3.48	1 / 0 / 0 / 0
	8	0.624 / 0.672 / 0.735 / 0.647	10.82 / 7.97 / 4.25 / 3.81	1 / 1 / 0 / 1
	16	0.621 / 0.741 / 0.742 / 0.665	11.05 / 3.71 / 3.43 / 4.66	2 / 0 / 0 / 1

Architecture	LoRA rank	Mean Dice (D / P / B / F)	Global IPE, % (D / P / B / F)	Seeds with a lost class (D / P / B / F)
	32	0.619 / 0.680 / 0.746 / 0.738	11.20 / 7.45 / 3.64 / 3.02	2 / 1 / 0 / 0
MobileSAM	2	0.741 / 0.805 / 0.740 / 0.808	5.35 / 1.55 / 5.32 / 1.30	1 / 0 / 1 / 0
	4	0.811 / 0.806 / 0.741 / 0.811	1.44 / 1.50 / 5.23 / 1.34	0 / 0 / 1 / 0
	8	0.811 / 0.813 / 0.813 / 0.810	1.53 / 1.60 / 1.48 / 1.32	0 / 0 / 0 / 0
	16	0.813 / 0.813 / 0.817 / 0.813	1.54 / 1.58 / 1.55 / 1.29	0 / 0 / 0 / 0
	32	0.674 / 0.817 / 0.818 / 0.687	9.23 / 1.58 / 1.46 / 1.76	2 / 0 / 0 / 1
MobileNet-UNet	2	0.416 / 0.413 / 0.549 / 0.410	17.98 / 19.88 / 12.40 / 2.61	3 / 3 / 3 / 3
	4	0.482 / 0.417 / 0.490 / 0.406	16.52 / 19.39 / 14.70 / 2.50	3 / 3 / 3 / 3
	8	0.485 / 0.414 / 0.544 / 0.412	15.48 / 19.79 / 13.96 / 2.60	2 / 3 / 2 / 3
	16	0.550 / 0.413 / 0.612 / 0.414	14.18 / 20.13 / 11.50 / 3.09	2 / 3 / 2 / 3
	32	0.359 / 0.486 / 0.418 / 0.476	20.83 / 15.96 / 19.54 / 5.83	3 / 3 / 3 / 3

S-III. Class-Specific Integrated-Phase Error and Dice

Table S-3 lists the class-specific IPE and Dice of the configurations of Table 1 and of the EdgeSAM loss ablation. A class that is not predicted in an image gives an IPE of 100% for that image, so the class-specific IPE is dominated by misclassification between morphologies; the global IPE over the whole foreground is the measure of integrated-phase preservation used in the main text.

TABLE S-3 Class-specific IPE (%) and Dice by mask value (1 discocyte, 2 spherocyte, 3 echinocyte, 4 stomatocyte), means over seeds 42, 43, and 44, including seeds in which a class was lost.

Configuration	Loss	IPE (%), 1	IPE (%), 2	IPE (%), 3	IPE (%), 4	Dice, 1	Dice, 2	Dice, 3	Dice, 4
MobileNet-UNet (*r*=8)	+ PMC + BGA + IPP (full)	79.5	100.0	28.2	100.0	0.891	0.000	0.756	0.000
MobileSAM (*r*=8)	+ PMC + BGA + IPP (full)	7.2	22.1	28.7	55.2	0.923	0.851	0.760	0.708
EdgeSAM (*r*=8)	+ PMC + BGA + IPP (full)	22.5	51.5	34.3	49.6	0.882	0.513	0.628	0.563
EdgeSAM (*r*=32)	+ PMC + BGA + IPP (full)	6.4	24.2	28.7	52.2	0.895	0.789	0.655	0.614
EdgeSAM (*r*=8)	Dice	6.5	64.2	28.7	49.4	0.893	0.343	0.668	0.592
EdgeSAM (*r*=8)	+ PMC	7.4	49.5	29.7	47.1	0.892	0.524	0.669	0.601
EdgeSAM (*r*=8)	+ PMC + BGA	7.6	22.1	31.5	46.7	0.892	0.783	0.654	0.612
EdgeSAM (*r*=32)	Dice	7.0	71.2	30.8	43.6	0.898	0.272	0.675	0.630
EdgeSAM (*r*=32)	+ PMC	6.5	48.4	28.0	47.8	0.896	0.522	0.678	0.625
EdgeSAM (*r*=32)	+ PMC + BGA	6.4	24.8	30.4	44.6	0.896	0.781	0.678	0.627

S-IV. Segmentation Accuracy versus Inference Throughput

Fig. S-3 relates the mean Dice of the rank-8 models to their FP32 forward-pass throughput (Table 4 of the main text); both quantities refer to rank 8. Dice was evaluated with the PyTorch checkpoints and is plotted at the same value for both runtimes.

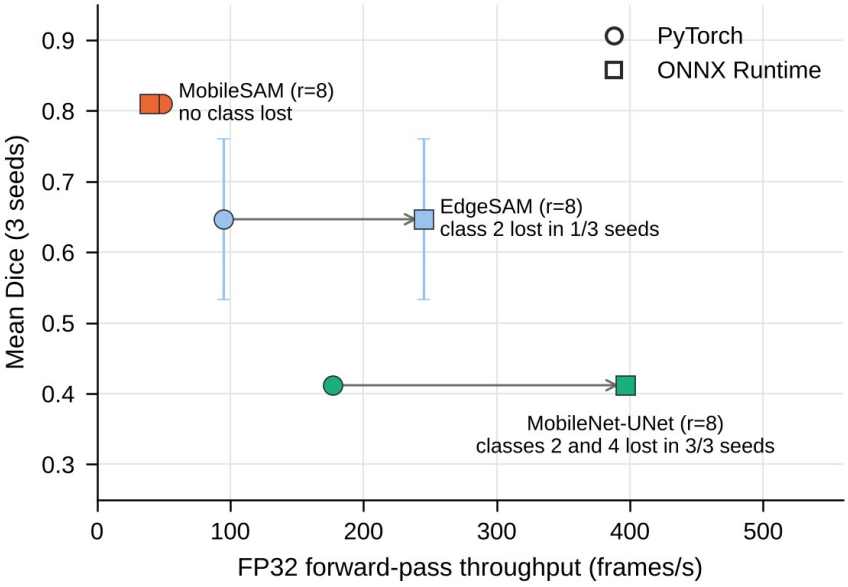


Fig. S-3 Mean Dice (mean ± SD over three seeds) against FP32 forward-pass throughput (mean of three timing passes) of the rank-8 models. Circles: PyTorch; squares: ONNX Runtime; class losses are noted next to each model.

S-V. Phase-Gradient Alignment of Predicted Contours

Fig. S-4 shows predicted cell contours on the phase-gradient magnitude for one validation window, as a qualitative illustration of the boundary cue used by the BGA term. Its quantitative effect is assessed by the loss ablation (Table 3 of the main text and Table S-2).

PLACEHOLDER: Fig. S-4 (phase-gradient alignment)

Generate with: python artifacts/code/make_data_figures.py

(needs data/processed; writes artifacts/figures/manuscript/FigS4_gradient_alignment.png)

Fig. S-4 Phase map, phase-gradient magnitude, and contours of EdgeSAM (r=32) and MobileSAM (r=8) (seed 42) on the gradient magnitude, for the validation window with the most annotated foreground.

S-VI. Representative Phase Images over the Storage Period

Fig. S-5 shows one validation phase map from each of five storage days, with the EdgeSAM (r=32) contours.

PLACEHOLDER: Fig. S-5 (phase images over storage)

Generate with: python artifacts/code/make_data_figures.py

(needs data/processed; writes artifacts/figures/manuscript/FigS5_storage_images.png)

Fig. S-5 Phase maps from storage days 0, 8, 19, 27, and 47 (first validation image of each day); bottom row: EdgeSAM (*r*=32) contours (seed 42).

S-VII. Storage-Period Measurements per Day

Tables S-4 and S-5 give the per-day measurements behind Fig. 8 for EdgeSAM (*r*=32) and MobileSAM (*r*=8). Cells are 4-connected foreground components of 15 to 200 µm² that do not touch an unannotated region; phase-wrapped cells are excluded. Annotation columns give the mean and, in parentheses, the cell-level SD; prediction columns give the mean over seeds of the per-day cell mean and, in brackets, the smallest and largest seed value. Predicted components are not cell counts. On days marked †, fewer than 0.6 kept components were found per annotated cell, and the predicted means describe only the components that passed the filters. Fig. S-6 shows the kept-component ratio and the fate of all predicted components per day.

TABLE S-4 Per-day measurements: annotation and EdgeSAM (*r*=32), seeds 42, 43, and 44.

Day	Annotated cells	Area, annotation (µm²)	Dry mass, annotation (pg)	Kept components (seeds 42 / 43 / 44)	Kept / annotated	Area, prediction (µm²)	Dry mass, prediction (pg)
0	185	52.1 (21.6)	31.7 (14.3)	142 / 146 / 154	0.80	50.5 [48.4–52.6]	31.5 [30.5–32.2]
5	211	51.2 (21.9)	33.2 (15.0)	139 / 144 / 159	0.70	58.5 [56.8–60.2]	37.3 [36.7–37.7]
8	220	51.9 (24.4)	34.9 (17.8)	119 / 114 / 112	0.52 †	59.3 [55.9–64.9]	39.3 [37.4–42.3]
12	230	54.9 (31.8)	36.3 (22.3)	103 / 108 / 124	0.49 †	61.6 [58.6–64.0]	40.2 [38.6–41.2]
15	206	59.3 (32.6)	40.9 (25.1)	51 / 53 / 74	0.29 †	60.7 [59.6–61.7]	41.6 [40.1–42.8]
19	162	49.8 (23.7)	32.2 (15.6)	124 / 124 / 130	0.78	54.5 [50.7–57.7]	35.5 [33.6–36.8]
23	108	49.5 (20.3)	30.6 (12.4)	92 / 95 / 95	0.87	49.8 [48.4–50.7]	31.3 [30.8–31.7]
27	93	50.8 (22.9)	33.0 (13.3)	84 / 85 / 89	0.92	49.9 [48.6–51.5]	33.2 [32.7–34.4]
30	188	46.0 (24.0)	33.5 (15.2)	160 / 164 / 161	0.86	44.5 [43.9–44.8]	32.8 [32.5–33.0]
37	157	36.7 (16.8)	31.0 (10.4)	144 / 153 / 156	0.96	35.5 [33.9–37.3]	30.1 [28.9–31.2]
47	228	27.1 (11.0)	28.8 (7.5)	180 / 192 / 183	0.81	28.9 [28.2–29.3]	30.1 [29.4–30.6]

TABLE S-5 Per-day measurements: annotation and MobileSAM (*r*=8), seeds 42, 43, and 44.

Day	Annotated cells	Area, annotation (µm²)	Dry mass, annotation (pg)	Kept components (seeds 42 / 43 / 44)	Kept / annotated	Area, prediction (µm²)	Dry mass, prediction (pg)
0	185	52.1 (21.6)	31.7 (14.3)	138 / 138 / 140	0.75	52.5 [52.0–53.3]	33.2 [32.9–33.8]
5	211	51.2 (21.9)	33.2 (15.0)	142 / 150 / 145	0.69	57.1 [56.1–58.4]	37.7 [37.2–38.5]
8	220	51.9 (24.4)	34.9 (17.8)	104 / 105 / 107	0.48 †	55.3 [54.8–55.8]	37.3 [36.9–37.6]
12	230	54.9 (31.8)	36.3 (22.3)	105 / 104 / 106	0.46 †	59.7 [58.4–61.0]	39.7 [39.0–40.6]
15	206	59.3 (32.6)	40.9 (25.1)	34 / 32 / 40	0.17 †	56.8 [51.2–62.7]	39.1 [35.7–42.8]
19	162	49.8 (23.7)	32.2 (15.6)	127 / 127 / 130	0.79	52.2 [51.0–54.0]	34.7 [34.0–35.5]
23	108	49.5 (20.3)	30.6 (12.4)	89 / 91 / 93	0.84	50.4 [49.4–51.6]	31.7 [31.3–32.1]
27	93	50.8 (22.9)	33.0 (13.3)	81 / 81 / 83	0.88	52.9 [52.6–53.2]	35.5 [35.2–35.7]
30	188	46.0 (24.0)	33.5 (15.2)	156 / 146 / 150	0.80	47.4 [45.9–48.2]	35.4 [34.8–35.8]
37	157	36.7 (16.8)	31.0 (10.4)	147 / 146 / 146	0.93	39.2 [38.8–39.6]	32.8 [32.4–33.2]
47	228	27.1 (11.0)	28.8 (7.5)	205 / 193 / 197	0.87	29.5 [28.0–30.6]	31.5 [30.5–32.3]

PLACEHOLDER: Fig. S-6 (predicted component counts)

Generate with: python artifacts/code/make_data_figures.py

(needs data/processed; writes artifacts/figures/manuscript/FigS6_component_fate.png)

Fig. S-6 Predicted components of EdgeSAM (r=32) per storage day: (a) kept components divided by annotated cells (markers: seeds); (b) fate of the predicted components inside the annotated field (mean of three seeds, wrapped components included), with the annotated cells kept by the same filters.

S-VIII. Date-Clustered Storage Statistics

Table S-6 gives, for each segmentation and training seed, the image-level Spearman correlation with storage day, the permutation p-value with the 11 acquisition dates permuted as whole blocks (10,000 permutations), the 95% date-bootstrap interval (2,000 resamples), and the range of the correlation when each date is left out, for projected area, dry mass, and the per-image difference between predicted and annotation-derived dry mass. With 11 clusters and an undocumented donor structure, these tests are approximate and exploratory.

TABLE S-6 Date-clustered storage statistics of the per-image medians (wrapped cells excluded).

Segmentation	Seed	Quantity (per-image median)	Spearman ρ	p (date permutation)	95% date-bootstrap interval	Leave-one-date-out range
Annotation	–	Area	−0.53	0.017	−0.85 to −0.004	−0.64 to −0.38
Annotation	–	Dry mass	−0.11	0.26	−0.32 to 0.12	−0.20 to −0.04
EdgeSAM (r=32)	42	Area	−0.75	0.0009	−0.93 to −0.27	−0.84 to −0.66
EdgeSAM (r=32)	43	Area	−0.60	0.008	−0.86 to −0.08	−0.68 to −0.48
EdgeSAM (r=32)	44	Area	−0.60	0.005	−0.87 to −0.11	−0.68 to −0.47
EdgeSAM (r=32)	42	Dry mass	−0.32	0.10	−0.56 to 0.07	−0.49 to −0.27
EdgeSAM (r=32)	43	Dry mass	−0.24	0.17	−0.46 to 0.05	−0.35 to −0.18
EdgeSAM (r=32)	44	Dry mass	−0.26	0.086	−0.48 to 0.03	−0.41 to −0.20
EdgeSAM (r=32)	42	Dry-mass difference (prediction − annotation)	−0.29	0.078	−0.45 to 0.06	−0.39 to −0.22
EdgeSAM (r=32)	43	Dry-mass difference (prediction − annotation)	−0.21	0.19	−0.43 to 0.07	−0.31 to −0.15
EdgeSAM (r=32)	44	Dry-mass difference (prediction − annotation)	−0.23	0.088	−0.42 to 0.05	−0.34 to −0.18
MobileSAM (r=8)	42	Area	−0.59	0.004	−0.86 to −0.08	−0.66 to −0.46
MobileSAM (r=8)	43	Area	−0.59	0.007	−0.87 to −0.10	−0.67 to −0.46
MobileSAM (r=8)	44	Area	−0.49	0.024	−0.82 to 0.02	−0.60 to −0.33
MobileSAM (r=8)	42	Dry mass	−0.14	0.36	−0.38 to 0.15	−0.27 to −0.08
MobileSAM (r=8)	43	Dry mass	−0.20	0.23	−0.45 to 0.13	−0.36 to −0.15
MobileSAM (r=8)	44	Dry mass	−0.16	0.34	−0.43 to 0.17	−0.32 to −0.09
MobileSAM (r=8)	42	Dry-mass difference (prediction − annotation)	−0.03	0.78	−0.19 to 0.15	−0.11 to 0.01
MobileSAM (r=8)	43	Dry-mass difference (prediction − annotation)	−0.05	0.63	−0.22 to 0.14	−0.14 to −0.008
MobileSAM (r=8)	44	Dry-mass difference (prediction − annotation)	−0.003	0.98	−0.22 to 0.22	−0.11 to 0.04

S-IX. Supplementary Data 1

Supplementary Data 1 (Excel workbook) lists, for EdgeSAM (r=32) and MobileSAM (r=8), every training seed and storage day: kept components, images, mean and cell-level SD of projected area, dry mass, and circularity, class shares (mask values 1 to 4), and the ratio of kept components to annotated cells; a second sheet gives the same quantities for the annotation. All values are computed from the per-cell tables of the corresponding runs; cells are not pooled across seeds.